BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 206 · Data structure selection challenge

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is an accumulator?

Q2. What is a collision?

Q3. What is a head pointer?

Q4. Which structure is used for Undo and why?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

LIFO — Last In, First Out.

FIFO — First In, First Out.

lookup — Retrieve a value using a key.

balanced BST — Tree kept at relatively small height.

trade-off — Benefit balanced against a drawback.

New learning

Identify the operation — The print job that arrived first must leave first.

Match the structure — Enqueue adds jobs; dequeue removes the oldest one. This is a queue.

Justify and qualify — A stack gives LIFO, which is the wrong order. Other systems may also need priority queues if priority overrides arrival order.

Worked example

A browser Back button must return to the most recently visited page. Choose a structure.

1. Predict — The last visited page is the first one that must be removed.

2. Explain — Use a stack: push visits and pop the most recent page.

We Do · Practise with guidance

A ticketing desk must serve requests in arrival order. Which structure?

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Tickets arrive: A → B → C. Serve one ticket.
```

Predict

The first request A must be the first one handled.

Explain

Use a queue, because FIFO removes the oldest item.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Select a structure for processing jobs in arrival order.

Question 2

Choose stack or hash table to retrieve customer records by unique ID. Justify.

Question 3

Compare a balanced BST and hash table for lookup and sorted output.

Exit Ticket · Answer before checking

A hash table is always O(1) even in the worst case.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A variable storing a running total.

Q2. Two keys map to the same initial bucket.

Q3. A reference to the first node.

Q4. A stack: the most recent operation must be undone first.

You Do pages · Independent questions

Q1. Queue (FIFO).

Q2. Hash table: hashing IDs supports expected fast exact-key retrieval; stacks expose only the latest item.

Q3. Balanced BST allows ordered traversal and O(log n) search; a hash table has expected O(1) lookup but does not inherently keep keys ordered.

Exit Ticket · Model answer

False. Collisions can make lookup much slower, sometimes O(n).

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: